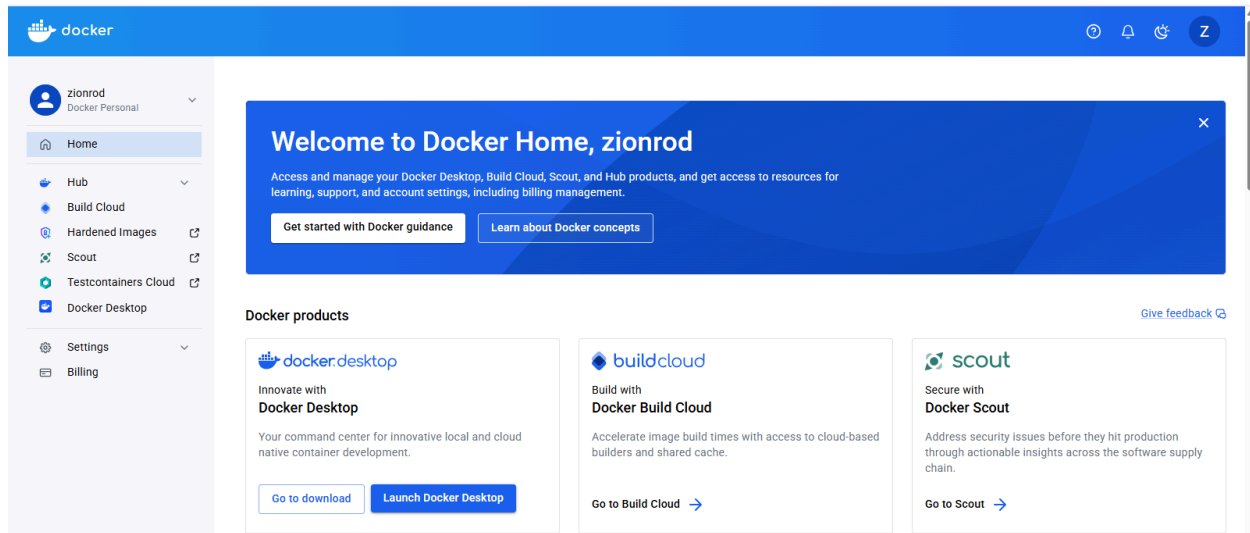


Creating a Docker Account



1. docker version

```
Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MOAD>docker --version
Docker version 27.2.0, build 3ab4256

C:\Users\MOAD>
```

2. docker login

```
C:\Users\MOAD>docker login
Authenticating with existing credentials...
Login did not succeed, error: error during connect: Post "http://%2F%2F.%2Fpipe%2FdockerDesktopLinuxEngine/v1.47/auth":
open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.

USING WEB BASED LOGIN
To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: MLFJ-SSTB
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...
Login Succeeded

C:\Users\MOAD>
```

3. docker images

```
C:\Users\MOAD>docker images
error during connect: Head "http://%2F%2F.%2Fpipe%2FdockerDesktopLinuxEngine/_ping": open //./pipe/dockerDesktopLinuxEng
ine: The system cannot find the file specified.
```

```
C:\Users\MOAD>docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
shreyal369/testimage 123                 0ddcfc4769f6       6 days ago         78.1MB
Shreya/testimage     123                 ef71cd865b94       6 days ago         78.1MB
ubuntu               latest             f794f40ddfff       5 weeks ago        78.1MB
mark/markg           latest             8c982b1546a4       5 months ago       78.1MB
<none>               <none>             eb2605a8bd15       5 months ago       78.1MB
ubuntu               <none>             97bed23a3497       6 months ago       78.1MB
```

4. docker pull image

```
C:\Users\MOAD>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\MOAD>|
```

6. docker images help

```
C:\Users\MOAD>docker images --help

Usage:  docker images [OPTIONS] [REPOSITORY[:TAG]]

List images

Aliases:
  docker image ls, docker image list, docker images

Options:
  -a, --all                Show all images (default hides intermediate images)
  --digests                Show digests
  -f, --filter filter       Filter output based on conditions provided
  --format string           Format output using a custom template:
                           'table':                Print output in table format
                           with column headers (default)
                           'table TEMPLATE':        Print output in table format
                           using the given Go template
                           'json':                  Print in JSON format
                           'TEMPLATE':              Print output using the given
                           Go template.
                           Refer to https://docs.docker.com/go/formatting/
                           for more information about formatting output with
                           templates
  --no-trunc               Don't truncate output
  -q, --quiet              Only show image IDs
  --tree                   List multi-platform images as a tree (EXPERIMENTAL)

C:\Users\MOAD>|
```

```
C:\Users\MOAD>docker images -a
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
shreya1369/testimage	123	0ddcfc4769f6	6 days ago	78.1MB
Shreya/testimage	123	ef71cd865b94	6 days ago	78.1MB
ubuntu	latest	f794f40ddfff	5 weeks ago	78.1MB
mark/markg	latest	8c982b1546a4	5 months ago	78.1MB
<none>	<none>	eb2605a8bd15	5 months ago	78.1MB
ubuntu	<none>	97bed23a3497	6 months ago	78.1MB

```
C:\Users\MOAD>docker images -q
0ddcfc4769f6
ef71cd865b94
f794f40ddfff
8c982b1546a4
eb2605a8bd15
97bed23a3497
```

```
C:\Users\MOAD>docker images -f "dangling=ture"
Error response from daemon: invalid filter 'dangling=[ture]'
```

```
C:\Users\MOAD>docker images -f "dangling=true"
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
<none>        <none>    eb2605a8bd15   5 months ago   78.1MB
```

```
C:\Users\MOAD>|
```

```
C:\Users\MOAD>docker images -f "dangling=false"
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
shreya1369/testimage  123       0ddcfc4769f6   6 days ago     78.1MB
Shreya/testimage    123       ef71cd865b94   6 days ago     78.1MB
ubuntu           latest    f794f40ddfff   5 weeks ago     78.1MB
mark/markg        latest    8c982b1546a4   5 months ago    78.1MB
ubuntu           <none>    97bed23a3497   6 months ago    78.1MB
```

```
C:\Users\MOAD>|
```

```
C:\Users\MOAD>docker images -f "dangling=false" -q
0ddcfc4769f6
ef71cd865b94
f794f40ddfff
8c982b1546a4
97bed23a3497
```

7. Docker run commands

```
C:\Users\MOAD>docker run ubuntu
```

```
C:\Users\MOAD>|
```

8. docker ps

```
C:\Users\MOAD>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ald42d5dac73dberg	ubuntu	"/bin/bash"	38 seconds ago	Exited (0) 37 seconds ago		affectionate_gol
49cfadc450f5	ubuntu	"bash"	6 days ago	Exited (255) 4 days ago		Shreya
49ea005d6deb	mark/markg	"bash"	5 months ago	Exited (137) 5 months ago		objective_solomo
6ead94008828	mark/markg	"bash"	5 months ago	Exited (137) 5 months ago		hopeful_mendel
18408e5cf9db	mark/markg:latest	"bash"	5 months ago	Exited (0) 5 months ago		xenodochial_eins
9c180f94c5f5	mark/markg	"bash"	5 months ago	Exited (137) 5 months ago		practical_nash
57ae0385a6aa	97bed23a3497	"/bin/bash"	5 months ago	Exited (137) 5 months ago		relaxed_hugle

```
C:\Users\MOAD>
```

The screenshot shows the Docker Desktop interface. At the top, there's a search bar and a 'Containers' tab. Below the tab, there are statistics for container CPU usage (0.00% / 1200%) and memory usage (880KB / 7.46GB). A search bar is present, and a toggle switch for 'Only show running containers' is visible. The main table lists several containers, with 'bidi_30' highlighted. The table columns are Name, Container ID, Image, Port(s), CPU (%), and Last started.

Name	Container ID	Image	Port(s)	CPU (%)	Last started
bidi_30	b15f85ce7e5a	ubuntu:<none>	-	0%	1 minute ago
bidi_container	cb4a6dbaff08	ubuntu:<none>	-	0%	4 days ago
chitra	5b09b146b4ef	ubuntu:<none>	-	0%	2 minutes ago
focused_swirlies	a6cd8f9e5b80	ubuntu:<none>	-	0%	5 days ago
Sarah	c622767d8783	ubuntu:<none>	-	0%	7 days ago
sad_voihard	5cbf75e4f3f8	ubuntu:<none>	-	0%	21 minutes ago

10. Docker pause container

```
C:\Users\MOAD>docker pause b15f85ce7e5a899e27edcfd335742b8599f91b416de3aefc14a1c5e09c040f19
b15f85ce7e5a899e27edcfd335742b8599f91b416de3aefc14a1c5e09c040f19
```

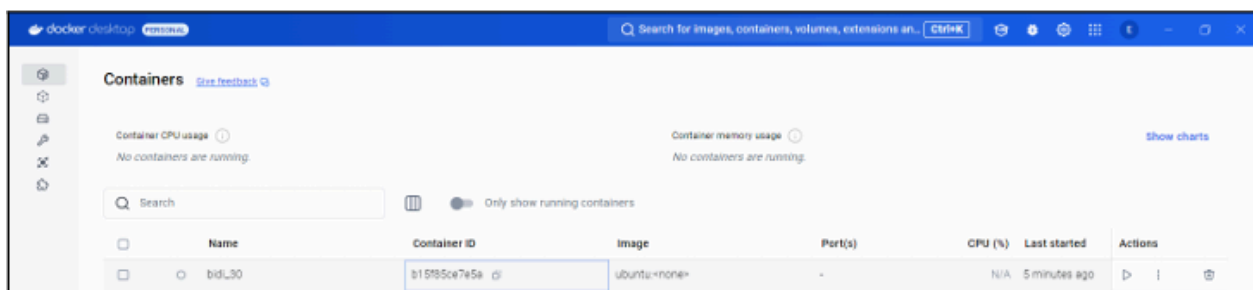
```
C:\Users\MOAD>
```



11. Docker stop container.

```
C:\Users\MOAD>docker stop b15f85ce7e5a899e27edcfd335742b8599f91b416de3aefc14a1c5e09c040f19
b15f85ce7e5a899e27edcfd335742b8599f91b416de3aefc14a1c5e09c040f19
```

```
C:\Users\MOAD>|
```



12. Docker top container

```
C:\Users\MOAD>docker top 5b09b146b4ef3fb6981dbfc0f7cd316ef51fb8b587f5900a83f5d53a9b94d19d
UID          PID          PPID         C             STIME        TTY          TIME          CMD
root          944          925          0             06:23        ?            00:00:00      bash
C:\Users\MOAD>|
```

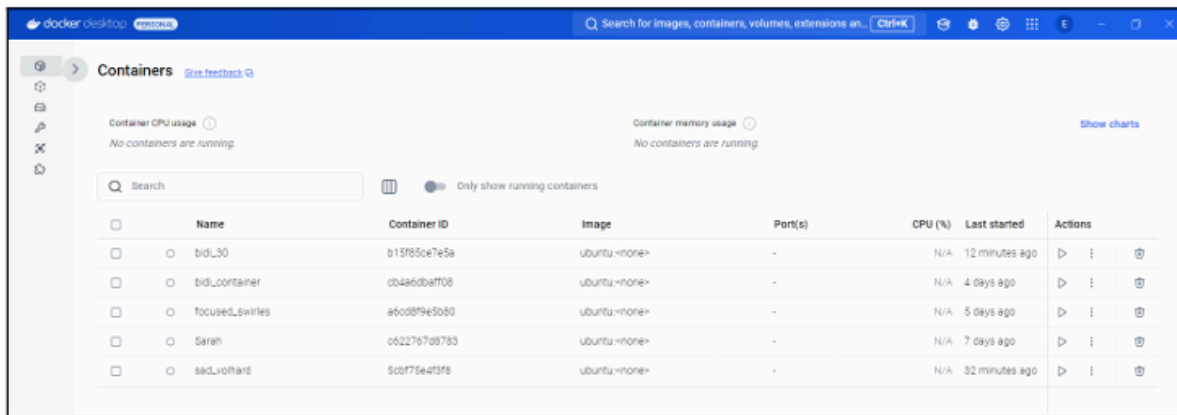
13. Docker stats container

```
C:\Users\MOAD> Docker stats
CONTAINER ID   NAME      CPU %       MEM USAGE / LIMIT   MEM %      NET I/O       BLOCK I/O  PIDS
5b09b146b4ef   chitra    0.00%       892KiB / 7.636GiB   0.01%      1.05kB / 0B   0B / 0B    1
CONTAINER ID   NAME      CPU %       MEM USAGE / LIMIT   MEM %      NET I/O       BLOCK I/O  PIDS
5b09b146b4ef   chitra    0.00%       892KiB / 7.636GiB   0.01%      1.05kB / 0B   0B / 0B    1
CONTAINER ID   NAME      CPU %       MEM USAGE / LIMIT   MEM %      NET I/O       BLOCK I/O  PIDS
5b09b146b4ef   chitra    0.00%       892KiB / 7.636GiB   0.01%      1.05kB / 0B   0B / 0B    1
```

14. Docker rm container

```
C:\Users\MOAD>docker rm 5b09b146b4ef3fb6981dbfc0f7cd316ef51fb8b587f5900a83f5d53a9b94d19d
5b09b146b4ef3fb6981dbfc0f7cd316ef51fb8b587f5900a83f5d53a9b94d19d
```

```
C:\Users\MOAD>|
```



15. Docker inspect

```
C:\Users\MOAD>docker inspect ubuntu
[
  {
    "Id": "sha256:186072bb1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c",
    "RepoTags": [
      "ubuntu:latest",
      "ubuntu:noble"
    ],
    "RepoDigests": [
      "ubuntu@sha256:186072bb1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c",
      "ubuntu@sha256:186072bb1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c"
    ],
    "Parent": "",
    "Comment": "",
    "Created": "2026-02-23T17:17:56.132176015Z",
    "DockerVersion": "27.2.0",
    "Author": "",
    "Config": {
      "Hostname": "",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
      ],
      "Cmd": [
        "/bin/bash"
      ],
      "Image": "",
      "Volumes": null,
      "WorkingDir": "",
      "Entrypoint": null,
      "OnBuild": null,
      "Labels": {
        "org.opencontainers.image.ref.name": "ubuntu",
        "org.opencontainers.image.version": "24.04"
      }
    }
  }
]
```

16. Docker rmi

```
C:\Users\MOAD>docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
bidishamahapatra/trialimage   new                4e594255652c       4 days ago         117MB
chitralata/trialimage        new                0d5f5f1a5d4d       4 days ago         117MB
chitralata              new                fac893a041a4       4 days ago         117MB
sarahm22/testimage          Sarah22            dcb4f3350aca       6 days ago         117MB
ubuntu                   resolute           91832dcd7bc5       3 weeks ago        157MB
ubuntu                   jammy-20260217    ce4a593b4e32       5 weeks ago        117MB
ubuntu                   latest            186072bba1b2       5 weeks ago        117MB
ubuntu                   noble             186072bba1b2       5 weeks ago        117MB
my_custom_image          latest            f4d91e044625       5 months ago       225MB
nginx                    latest            1beed3ca46ac       5 months ago       225MB

C:\Users\MOAD>docker rmi 91832dcd7bc5
Untagged: ubuntu:resolute
Deleted: sha256:91832dcd7bc5e44c098ecef0a251a5c5d596dae494b33fb248e01b6840f8ce0

C:\Users\MOAD>docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
bidishamahapatra/trialimage   new                4e594255652c       4 days ago         117MB
chitralata/trialimage        new                0d5f5f1a5d4d       4 days ago         117MB
chitralata              new                fac893a041a4       4 days ago         117MB
sarahm22/testimage          Sarah22            dcb4f3350aca       6 days ago         117MB
ubuntu                   jammy-20260217    ce4a593b4e32       5 weeks ago        117MB
ubuntu                   latest            186072bba1b2       5 weeks ago        117MB
ubuntu                   noble             186072bba1b2       5 weeks ago        117MB
my_custom_image          latest            f4d91e044625       5 months ago       225MB
nginx                    latest            1beed3ca46ac       5 months ago       225MB

C:\Users\MOAD>
```

17. Docker commit

```
C:\Users\MOAD>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
5cbf75e4f3f8   ubuntu   "/bin/bash"   36 minutes ago   Exited (0) 36 minutes ago   sad_volhard
b15f85ce7e5a   ubuntu   "bash"       4 days ago      Exited (137) 11 minutes ago   bidi_30
cb4a6dbaff08   ubuntu   "bash"       4 days ago      Exited (137) 4 days ago      bidi_container
a6cd8f9e5b80   ubuntu   "/bin/bash"   4 days ago      Exited (0) 4 days ago      focused_swirles
c622767d8783   ubuntu   "bash"       6 days ago      Exited (255) 4 days ago      Sarah

C:\Users\MOAD>docker commit a6cd8f9e5b80
sha256:3bcbbaa556fc9a219f58c58e130f501dc877d2b358fbf90fdbfb3f055f8669597

C:\Users\MOAD>
```

18. Docker push

```
C:\Users\MOAD>docker login
Authenticating with existing credentials...
Login Succeeded

C:\Users\MOAD>docker tag ubuntu:noble evonne/ubuntul

C:\Users\MOAD>docker push evonne/ubuntul
Using default tag: latest
The push refers to repository [docker.io/evonne/ubuntul]
817807f3c64e: Unavailable
server message: insufficient_scope: authorization failed

C:\Users\MOAD>docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
<none>              <none>             3bcbbaa556fc9      22 minutes ago     117MB
bidishamahapatra/trialimage   new                4e594255652c       4 days ago         117MB
chitralata/trialimage        new                0d5f5f1a5d4d       4 days ago         117MB
chitralata              new                fac893a041a4       4 days ago         117MB
sarahm22/testimage          Sarah22            dcb4f3350aca       6 days ago         117MB
ubuntu                   jammy-20260217    ce4a593b4e32       5 weeks ago        117MB
ubuntu                   latest            186072bba1b2       5 weeks ago        117MB
ubuntu                   noble             186072bba1b2       5 weeks ago        117MB
evonne/ubuntul          latest            186072bba1b2       5 weeks ago        117MB
evonne/ubuntul31         latest            186072bba1b2       5 weeks ago        117MB
my_custom_image          latest            f4d91e044625       5 months ago       225MB
nginx                    latest            1beed3ca46ac       5 months ago       225MB

C:\Users\MOAD>
```


	evonne/ubuntu1	latest	186072bba1b2	1 month ago	117.33 MB			
	evonne/ubuntu31	latest	186072bba1b2	1 month ago	117.33 MB			
	ubuntu	latest	186072bba1b2	1 month ago	117.33 MB			
	ubuntu	noble	186072bba1b2	1 month ago	117.33 MB			

19. Docker history image

```
C:\Users\MOAD>docker images -q
3bcbaa556fc9
4e594255652c
0d5f5f1a5d4d
fac893a041a4
dcb4f3350aca
ce4a593b4e32
186072bba1b2
186072bba1b2
f4d91e044625
1beed3ca46ac

C:\Users\MOAD>docker history ce4a593b4e32
IMAGE          CREATED        CREATED BY          SIZE      COMMENT
ce4a593b4e32   5 weeks ago   /bin/sh -c #(nop)  CMD ["/bin/bash"]   0B
<missing>      5 weeks ago   /bin/sh -c #(nop)  ADD file:87202021c36509f80... 87.5MB
<missing>      5 weeks ago   /bin/sh -c #(nop)  LABEL org.opencontainers... 0B
<missing>      5 weeks ago   /bin/sh -c #(nop)  LABEL org.opencontainers... 0B
<missing>      5 weeks ago   /bin/sh -c #(nop)  ARG LAUNCHPAD_BUILD_ARCH    0B
<missing>      5 weeks ago   /bin/sh -c #(nop)  ARG RELEASE                0B

C:\Users\MOAD>
```

a. Answer the following:

1. What are Docker containers and Docker images?

- Docker Image:

A Docker image is a read-only template used to create containers. It includes the application code, libraries, dependencies, and environment required to run an application.

- Docker Container:

A Docker container is a running instance of a Docker image. It is lightweight, isolated, and portable, allowing applications to run consistently across different environments.

2. Explain Docker architecture with diagram

Docker Architecture consists of three main components:

1. Docker Client

- User interface to interact with Docker
- Runs commands like docker run, docker build

2. Docker Host (Docker Engine)

- Core system that manages containers
- Includes:
 - Docker Daemon
 - Containers
 - Images

3. Docker Registry (Docker Hub)

- Stores Docker images

- Public or private repositories

Working Flow: ● Client sends command → Docker Daemon executes → Pulls image from Docker Hub → Runs container.

